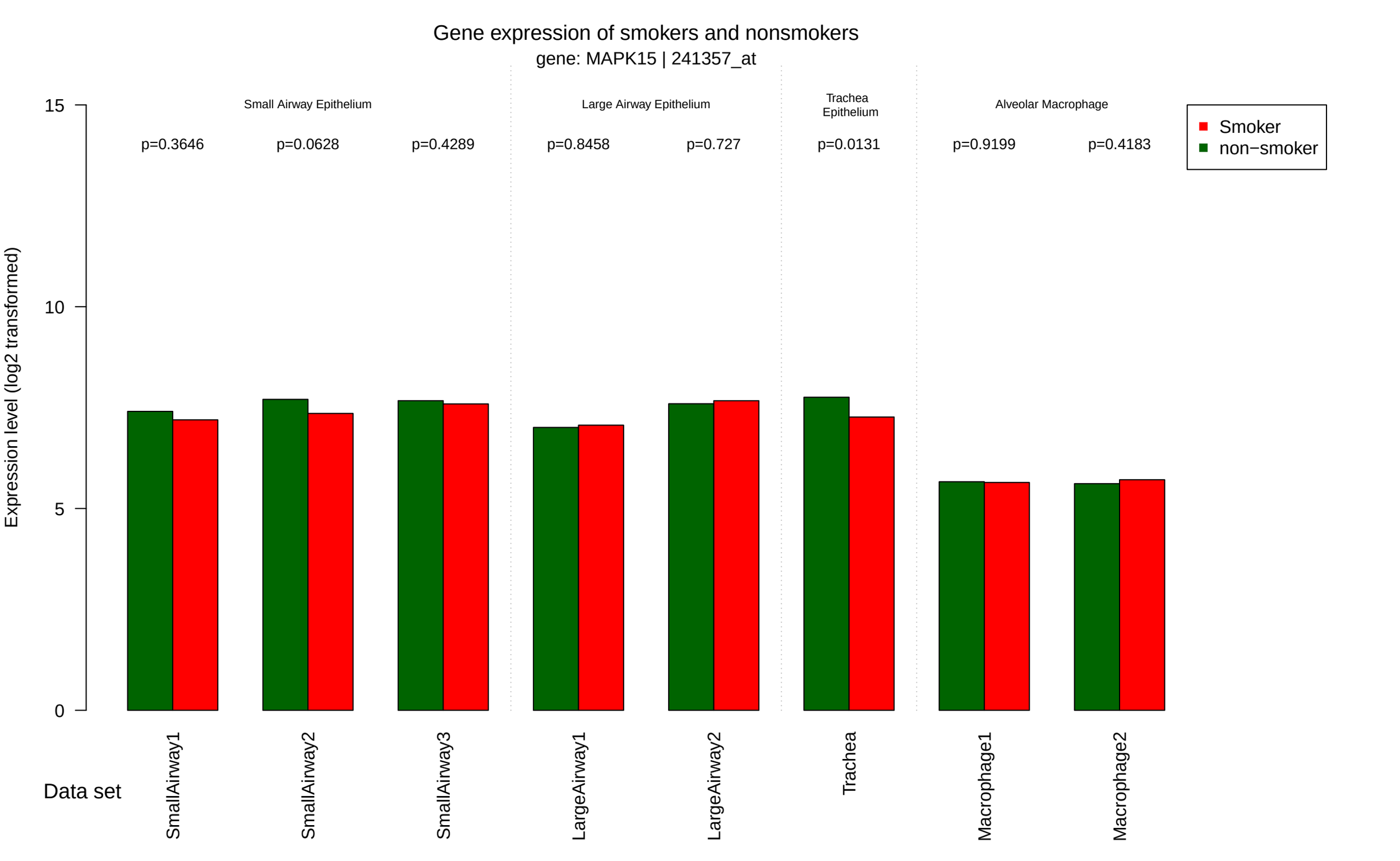
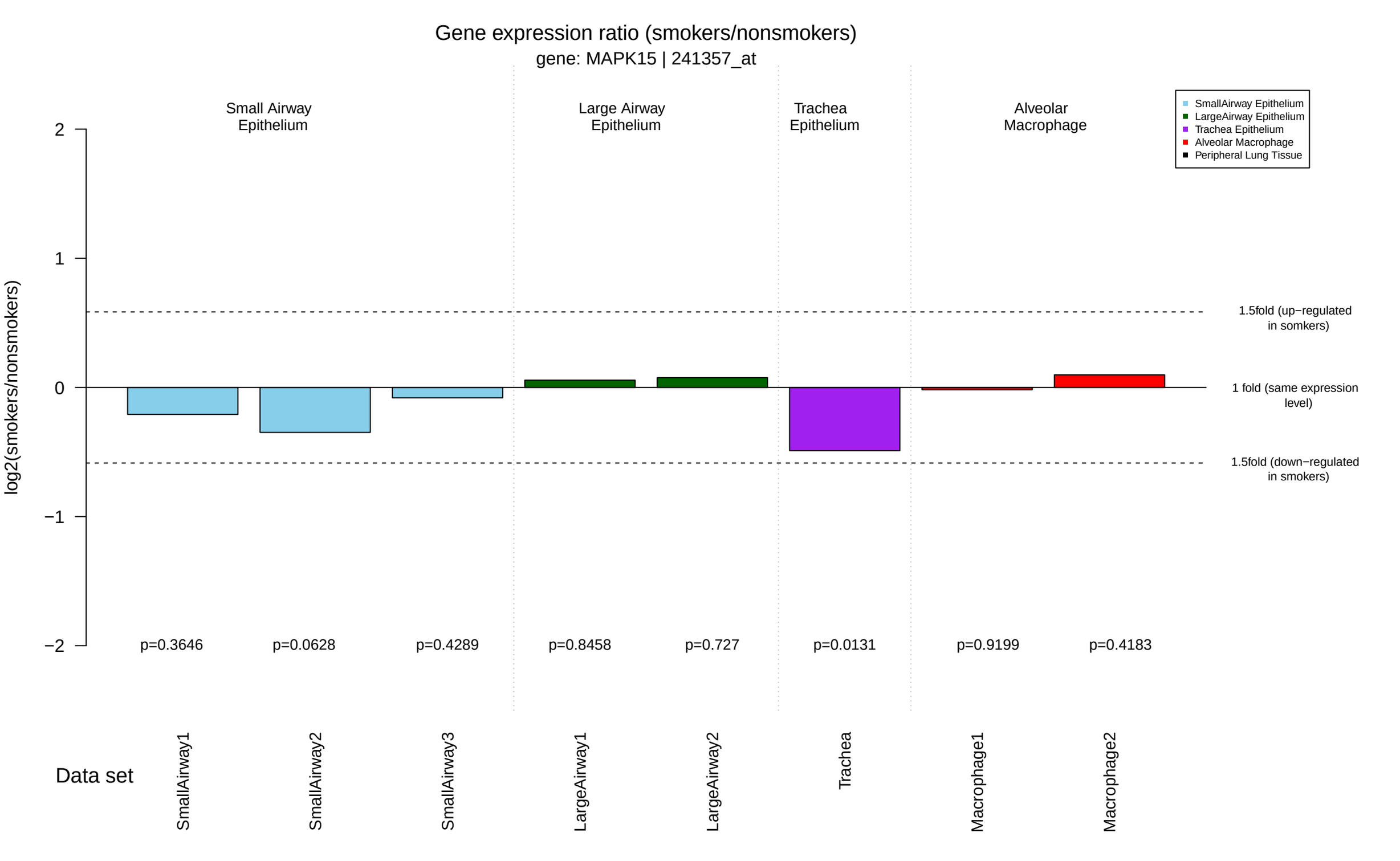
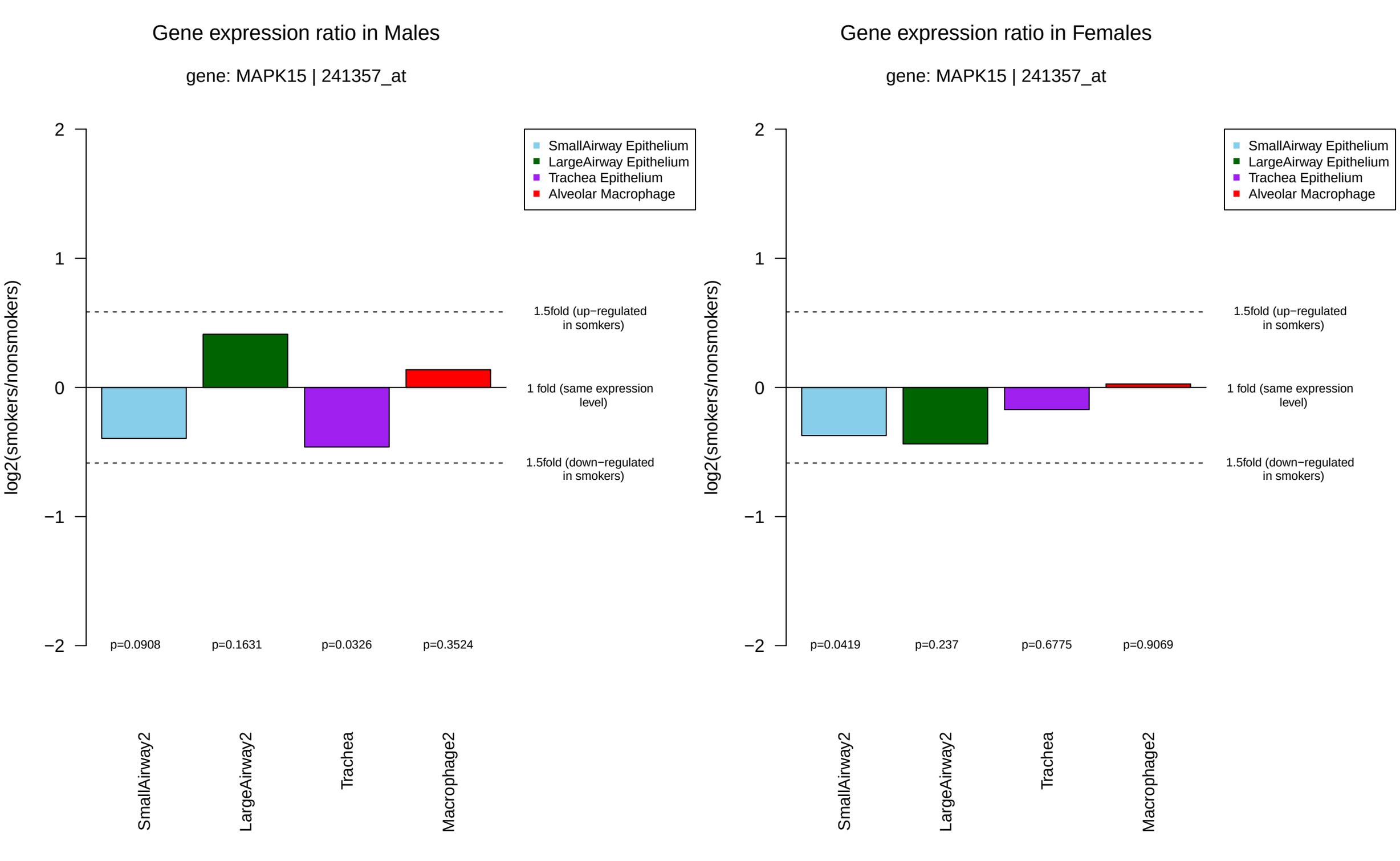
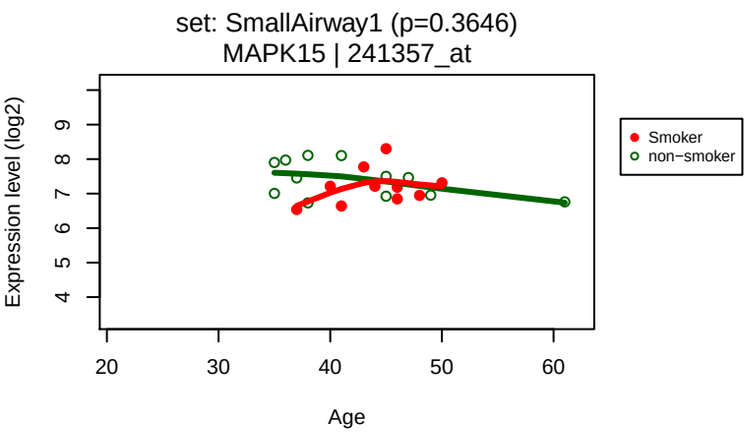
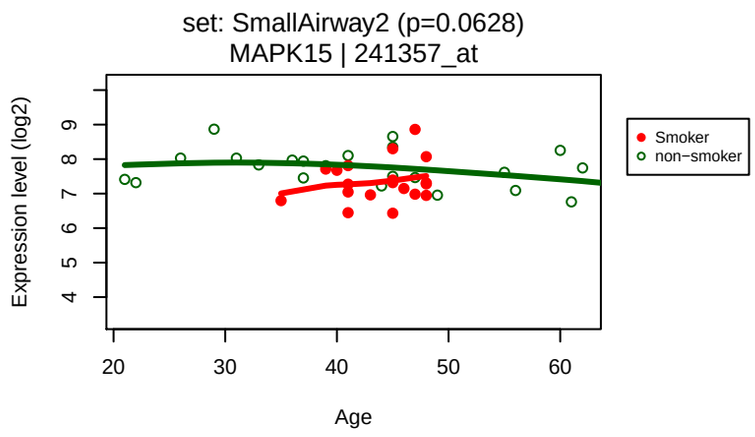
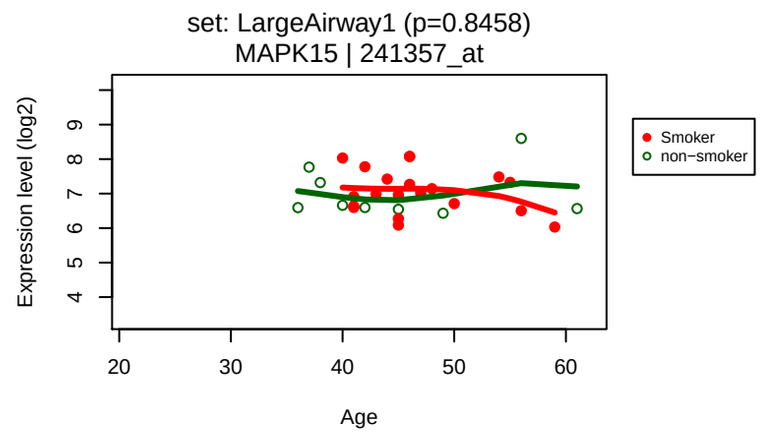

Gene expression of smokers and nonsmokers
gene: MAPK15 | 241357_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3646
p=0.0628
p=0.4289
p=0.8458
p=0.727
p=0.0131
p=0.9199
p=0.4183
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

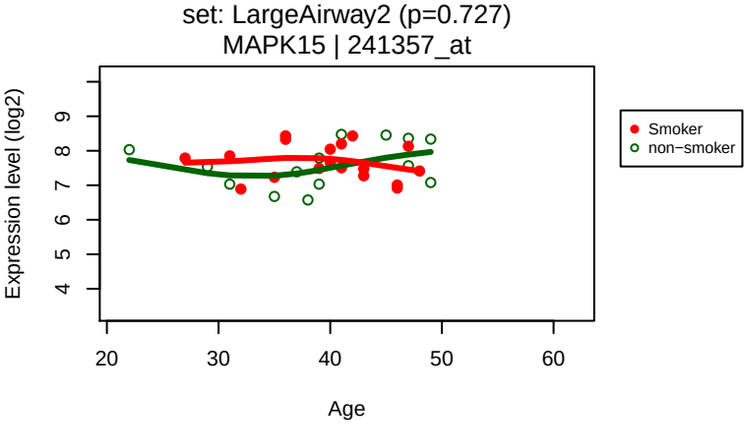

Gene expression ratio (smokers/nonsmokers)
gene: MAPK15 | 241357_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3646
p=0.0628
p=0.4289
p=0.8458
p=0.727
p=0.0131
p=0.9199
p=0.4183
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

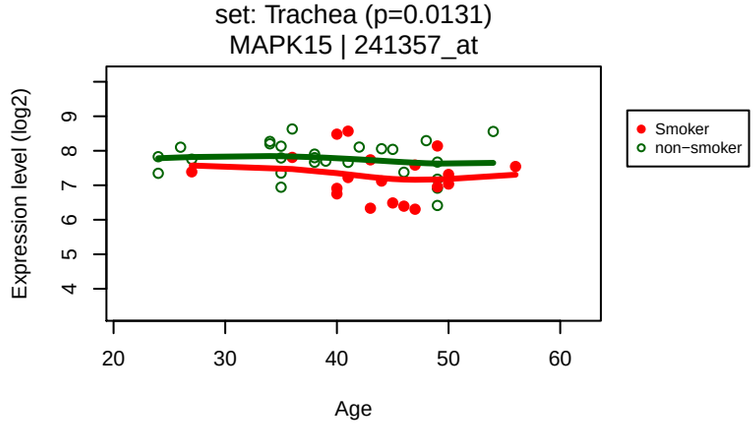

Gene expression ratio in Males
gene: MAPK15 | 241357_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0908
p=0.1631
p=0.0326
p=0.3524
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAPK15 | 241357_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0419
p=0.237
p=0.6775
p=0.9069
SmallAirway2
LargeAirway2
Trachea
Macrophage2

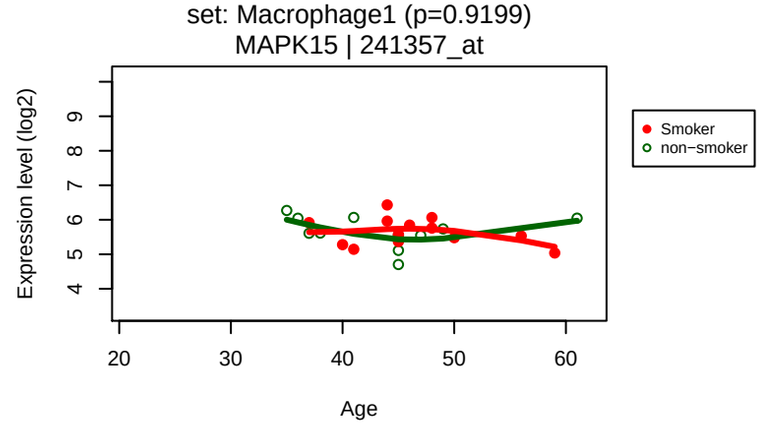

set: SmallAirway1 (p=0.3646)
MAPK15 | 241357_at
Expression level (log2)
Age
Smoker
non−smoker

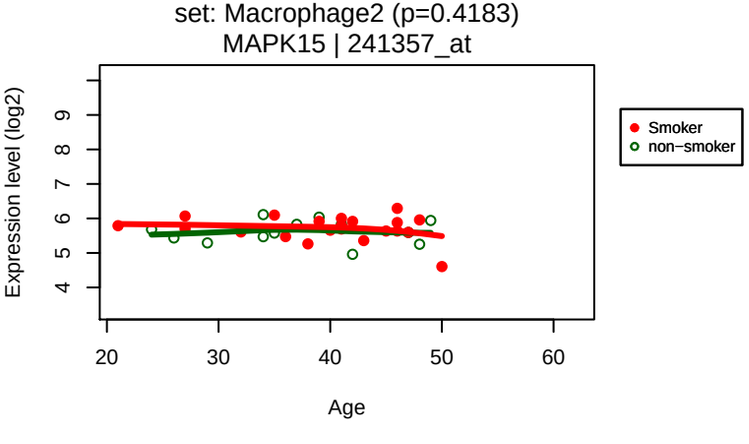

set: SmallAirway2 (p=0.0628)
MAPK15 | 241357_at
Expression level (log2)
Age
Smoker
non−smoker

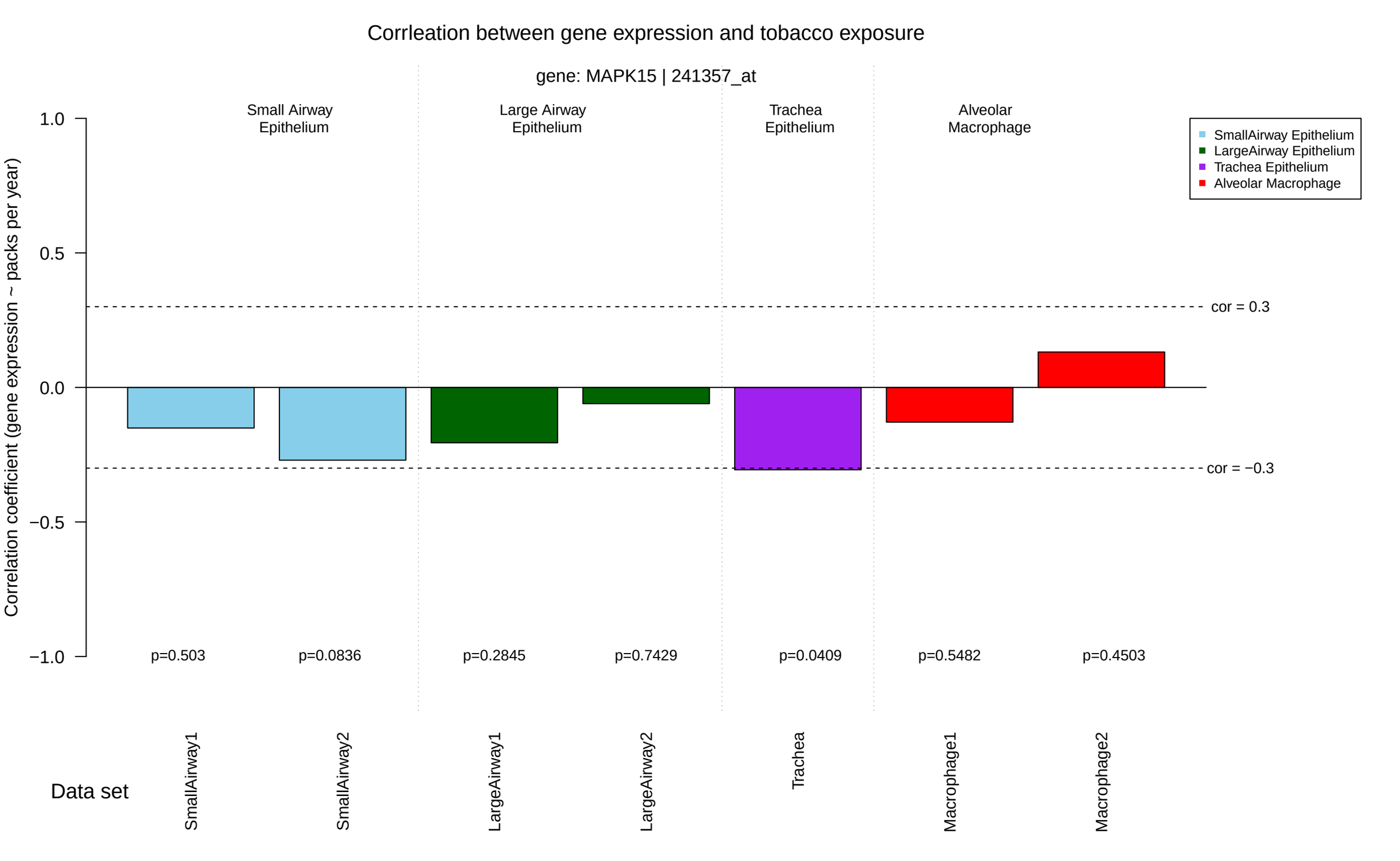

set: LargeAirway1 (p=0.8458)
MAPK15 | 241357_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.727)
MAPK15 | 241357_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0131)
MAPK15 | 241357_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9199)
MAPK15 | 241357_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4183)
MAPK15 | 241357_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAPK15 | 241357_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.503
p=0.0836
p=0.2845
p=0.7429
p=0.0409
p=0.5482
p=0.4503
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2